Pneumonia Symptoms and Diagnosis

Pneumonia is an infection that inflames your lungs' air sacs (alveoli). The air sacs may fill up with fluid or pus, causing symptoms such as a cough, fever, chills and trouble breathing.

What Are the Symptoms of Pneumonia?

Pneumonia symptoms can vary from so mild you barely notice them, to so severe that hospitalization is required. How your body responds to pneumonia depends on the type of germ causing the infection, your age and your overall health.

Questions about your symptoms?

Talk to our experts at the American Lung Association Lung HelpLine. Our service is free and we are here to help you by phone, web chat or email.

TALK TO THE LUNG HELPLINE (HTTPS://LUNG.ORG/HELP-SUPPORT/LUNG-HELPLINE)

The signs and symptoms of pneumonia may include:

- Cough (https://lung.org/lung-health-diseases/warning-signs-of-lung-disease/cough), which may produce greenish, yellow or even bloody mucus
- Fever

- Sweating and chills
- Shortness of breath (https://lung.org/lung-health-diseases/warning-signs-of-lung-disease/shortness-of-breath)
- Rapid breathing
- Chest pain when you breath or cough
- Loss of appetite, low energy and fatigue
- Nausea and vomiting, especially in small children
- Confusion, especially in older people

Bacterial Pneumonia Symptoms

Bacterial pneumonia, which is the most common form, tends to be more serious than other types of pneumonia (https://lung.org/lung-health-diseases/lung-disease-lookup/pneumonia/what-causes-pneumonia), with symptoms that require medical care. The symptoms of bacterial pneumonia can develop gradually or suddenly. Fever may rise as high as a dangerous 105 degrees F, with sweating, chills, rapid breathing and pulse rate. Lips and nailbeds may have a bluish color due to lack of oxygen in the blood. Older adults may

experience a change in mental state including confusion or low alertness. Mycoplasma pneumonia has different symptoms, including a severe cough.

Viral Pneumonia Symptoms

The symptoms of viral pneumonia usually develop over a period of several days. Early symptoms are similar to influenza symptoms (https://lung.org/lung-health-diseases/lung-disease-lookup/influenza/symptoms-causes-and-risk): fever, a dry cough, headache, muscle pain, and weakness. Within a day or two, the symptoms typically get worse, with increasing cough, shortness of breath and muscle pain. There may be a high fever and there may be blueness of the lips.

Symptoms in Certain Populations

Symptoms may vary in certain populations. Newborns and infants may not show any signs of the infection. Or, they may vomit, have a fever and cough, or appear restless, sick, or tired and without energy. Older adults and people with weakened immune systems may have fewer and milder symptoms, however symptoms associated with an underlying medical condition may worsen. Older adults may even have a lower-than-normal temperature and changes in mental awareness. For individuals that already have a chronic lung disease, those symptoms may worsen.

When to Call a Doctor

If you think you or your child has symptoms of pneumonia, don't wait for the disease to get even worse before you seek care. Call your doctor. And see your doctor right away if you have difficulty breathing, develop a bluish color in your lips and fingertips, have chest pain, a high fever, or a cough with mucus that is severe or is getting worse.

It's especially important to get medical attention for pneumonia if you are in a high-risk group, including adults older than age 65, children age two or younger, people with an underlying health condition or

weakened immune system. For some of these vulnerable individuals, pneumonia can quickly become a life-threatening condition.

How Is Pneumonia Diagnosed?

Sometimes pneumonia can be difficult to diagnose because the symptoms are so variable, and are often very similar to those seen in a cold or influenza. To diagnose pneumonia, and to try to identify the germ that is causing the illness, your doctor will ask questions about your medical history, do a physical exam, and run some tests.

Medical History

Your doctor will ask you questions about your signs and symptoms, and how and when they began. To help figure out if your infection is caused by bacteria, viruses or fungi, you may be asked some questions about possible exposures, such as:

- Any recent travel
- Your occupation
- Contact with animals
- Exposure to other sick people at home, work or school
- Whether you have recently had another illness

Physical Exam

Your doctor will listen to your lungs with a stethoscope. If you have pneumonia, your lungs may make crackling, bubbling, and rumbling sounds when you inhale.

Diagnostic Tests

If your doctor suspects you may have pneumonia, they will probably recommend some tests to confirm the diagnosis and learn more about your infection. These may include:

- **Blood tests** to confirm the infection and to try to identify the germ that is causing your illness.

- **Chest X-ray** to look for the location and extent of inflammation in your lungs.
- **Pulse oximetry** (https://lung.org/lung-health-diseases/lung-procedures-and-tests/pulse-oximetry) to measure the oxygen level in your blood. Pneumonia can prevent your lungs from moving enough oxygen into your bloodstream.
- **Sputum test** on a sample of mucus (sputum) taken after a deep cough, to look for the source of the infection.

If you are considered a high-risk patient because of your age and overall health, or if you are hospitalized, the doctors may want to do some additional tests, including:

- **CT scan** (https://lung.org/lung-health-diseases/lung-procedures-and-tests/ct-scan) of the chest to get a better view of the lungs and look for abscesses or other complications.
- **Arterial blood gas test**, to measure the amount of oxygen in a blood sample taken from an artery, usually in your wrist. This is more accurate than the simpler pulse oximetry.
- **Pleural fluid culture**, which removes a small amount of fluid from around tissues that surround the lung, to analyze and identify bacteria causing the pneumonia.
- **Bronchoscopy** (https://lung.org/lung-health-diseases/lung-procedures-and-tests/bronchoscopy), a procedure used to look into the lungs' airways. If you are hospitalized and your treatment is not working well, doctors may want to see whether something else is affecting your airways, such as a blockage. They may also take fluid samples or a biopsy of lung tissue.

Pneumococcal pneumonia can disrupt your life for weeks

Pneumococcal vaccination is recommended for all adults 50+ and adults 19-49 with certain chronic medical conditions like asthma, COPD, diabetes and heart disease. In partnership with Pfizer.

LEARN MORE (HTTPS://LUNG.ORG/LUNG-HEALTH-DISEASES/LUNG-DISEASE-LOOKUP/PNEUMONIA/PNEUMOCOCCAL/HCP/HCP-RESOURCES)

Page last updated: August 14, 2025